Austo Automobiles

Business Analysis Report

–Aryaman Modi–

EXECUTIVE SUMMARY

This report looks at customer data from Austo Automobiles to understand how people are buying cars, who they are, and what kind of vehicles they prefer. The dataset includes information on 1,581 individuals — covering their age, income, marital status, profession, education, loan status, and car choices.

After exploring the data, we found that most customers are young, salaried professionals who tend to buy mid-range vehicles like hatchbacks and sedans. Female customers, although fewer in number, spend more on average than male customers. People who take personal loans are also responsible for a significant portion of the total money spent on car purchases. Surprisingly, whether or not someone's partner is working doesn't seem to affect how much they spend.

These findings give us clear opportunities for the business — like targeting salaried customers with sedan offers, promoting premium vehicles to female buyers, and making loan-based purchases easier for those who need financing.

ABOUT THE DATA

This dataset was shared by Austo Automobiles and contains information about 1,581 people who have purchased vehicles. Each row represents one person, and the columns include things like their age, gender, profession, salary, education, whether they took a loan, and what kind of car they bought.

The goal of working with this dataset is to find out what kind of people buy which types of cars, how much they usually spend, and what factors (like income or loan status) might influence their decisions. It includes both numbers (like salary and car price) and categories (like profession or car make).

Overall, the dataset is pretty clean and straightforward. It gives enough information to understand customer behavior and explore useful insights that could help the company make better decisions in marketing and sales.

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(1581, 14)
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DATA OVERVIEW

Before I started analyzing anything, I wanted to take a proper look at the data to understand what I'm working with.

There are **1,581 rows** in this dataset, and each row is about one person. There are **14 columns**, and they contain information like the person's age, gender, job, salary, the kind of car they bought, whether they took any loans, and so on.

What I found while checking the data:

- There were **no duplicate rows**, which means every person in the dataset is unique.

- There are **some missing values**, but only in two columns:

    - Gender : 53 people didn't have this info filled in.

    - Partner_Salary : missing in 106 rows.

    - The rest of the columns have **complete data**, so that's good.
      The data types look fine — for example, numbers are stored as numbers, and categories (like "Male" or "Salaried") are stored as text.

Small fixes I made:

- In the Gender column, there were a few typos like "Femal" and "Femle". I corrected them to "Female" to keep everything consistent.

- The Total_salary column adds up the person's salary and their partner's salary. I checked a few rows and it seems accurate, so that's a good sign.

Final thoughts on the data:

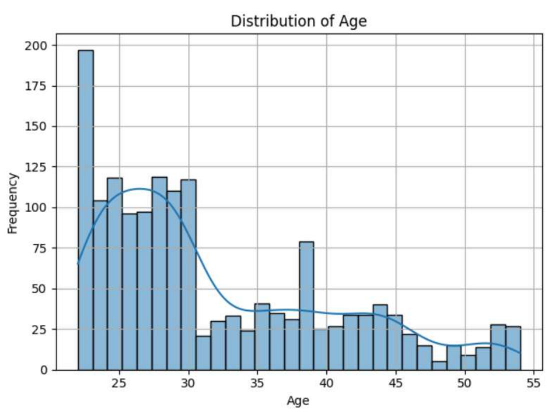
The dataset is actually quite clean. There weren't too many problems, and the small ones were easy to fix. Now it's ready to be analyzed so I can start looking for patterns and insights.

UNIVARIATE ANALYSIS (BOTH TYPES)

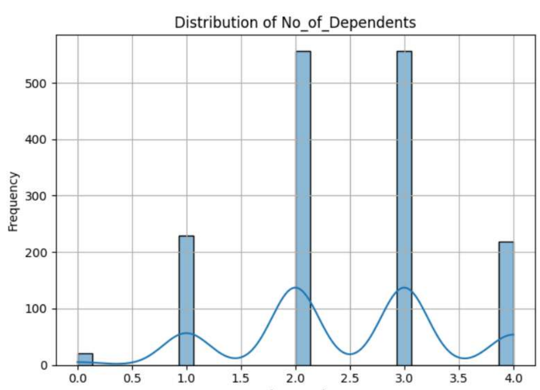
This section examines each variable individually to understand their main characteristics.

Numerical Variables:

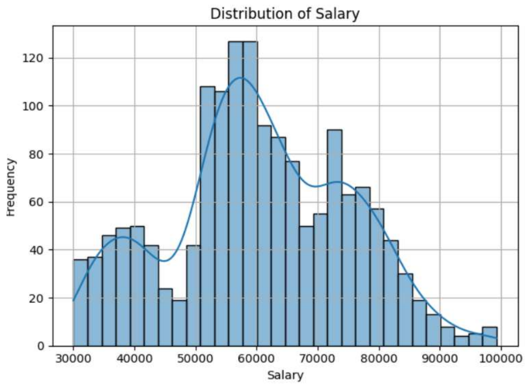
- **Age** ranges from 22 to 54 years, with an average around 32 years.



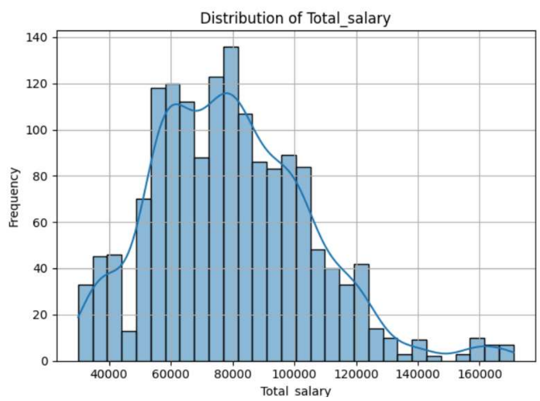
- Most individuals have about two to three dependents.



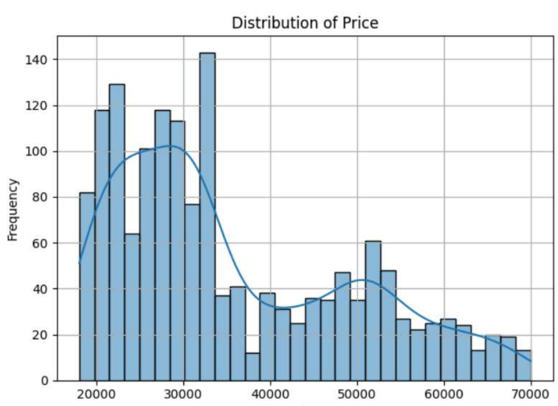
- Individual salaries vary from $30,000 up to nearly $100,000, with many people earning around $60,000.



- Partner salaries range widely, including many with no reported income and some earning up to $80,000.
- Total household income combines these, ranging from $30,000 to over $170,000.

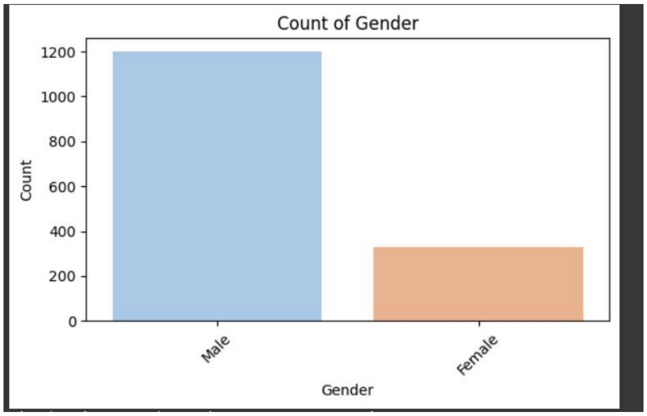


- Prices vary from $18,000 to $70,000, with an average near $35,600.

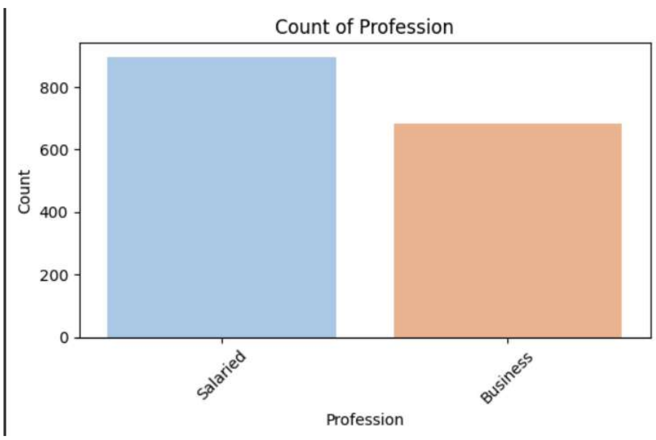


Categorical Variables:

- The dataset includes more males than females, with a small number of missing gender entries.

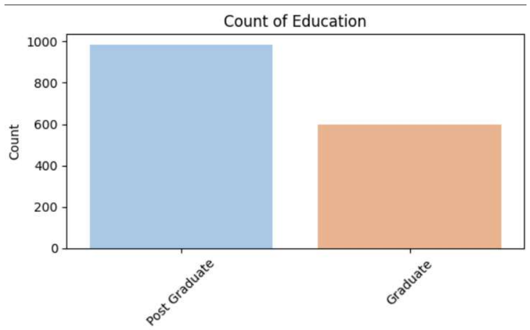


- There are more salaried employees than business owners.

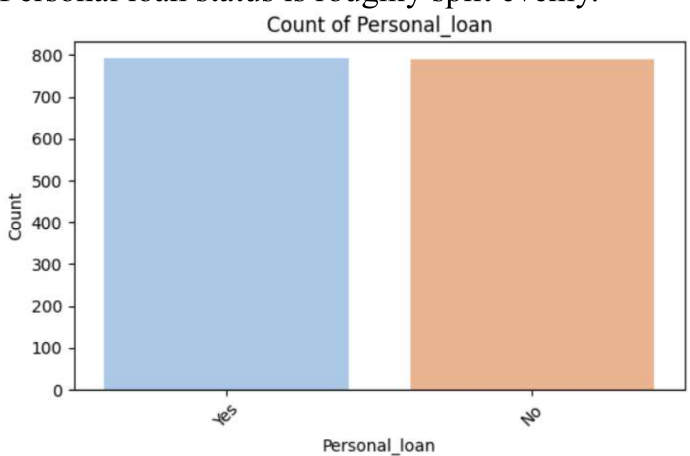


- Most individuals are married.

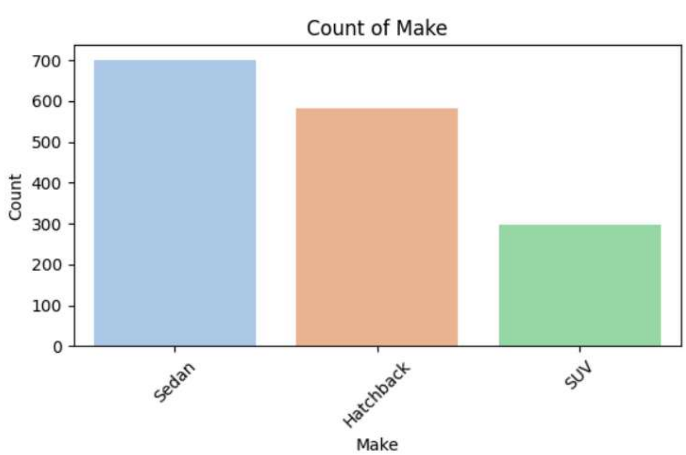
- Postgraduate education is more common than graduate education.



- Personal loan status is roughly split evenly.



- More people do not have a house loan compared to those who do.
- Slightly more partners are working than not.
- Among vehicle types, Sedans are the most common, followed by Hatchbacks and then SUVs.

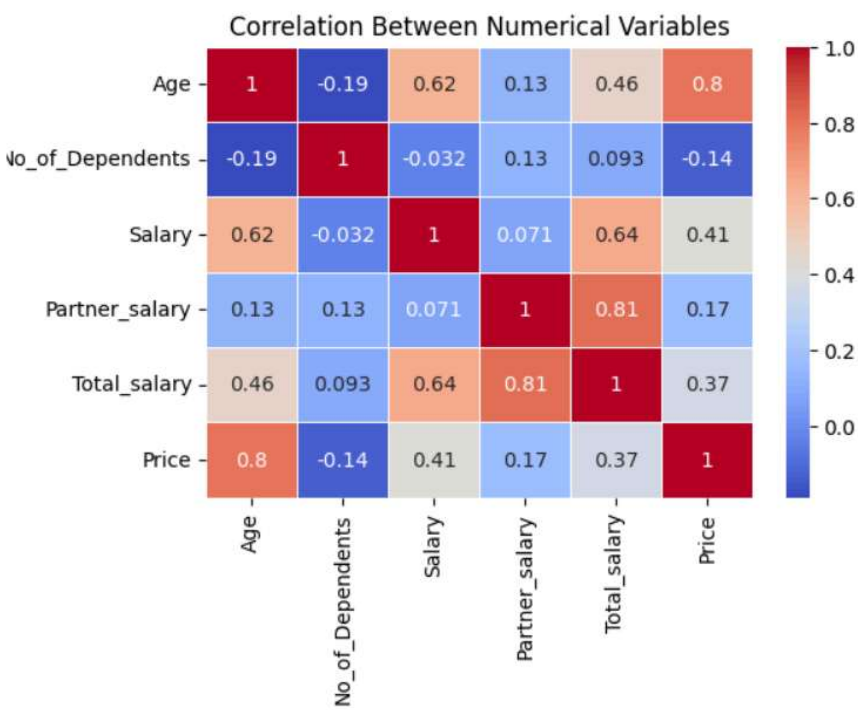


BIVARIATE ANALYSIS

Correlation Between Numerical Variables

To get a sense of how the numerical variables relate to each other, a correlation heatmap was used. As expected, there's a strong correlation between Salary and Total Salary, since total income includes both the individual's and the partner's salary. There's also a noticeable link between Total Salary and Price, which suggests that people with higher combined incomes tend to go for higher-priced options. Other relationships, like those between Age, Dependents, and Partner Salary with Price, appear to be much weaker, meaning they probably don't have a big direct impact.

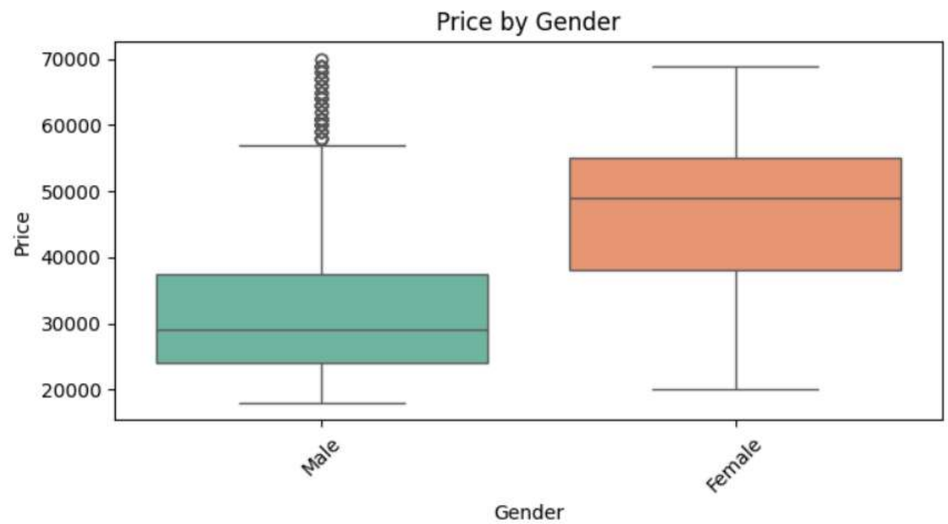
The heatmap gives a quick overview of which variables move together and which ones don't, helping narrow down what's worth focusing on in deeper analysis.



The heatmap shows how each numerical variable is related to the others. The strongest relationship is between Salary and Total Salary, which makes sense since Total Salary includes the individual's earnings. There's also a moderate correlation between Total Salary and Price, suggesting that people with higher household incomes might spend more. Most other relationships, such as Age with Salary or Dependents with Price, show weak or no meaningful correlation. This tells us that while income is clearly connected to spending or purchase price, other factors like age or family size don't have as strong a direct influence in this dataset.

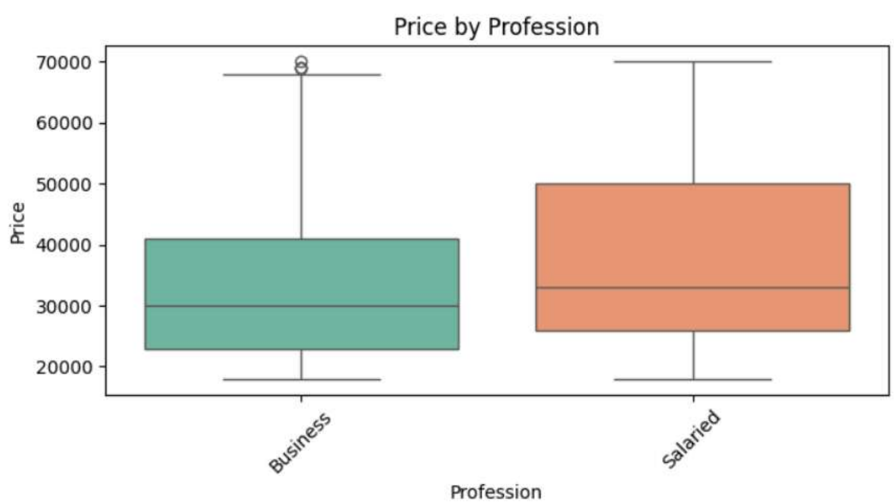
Categorical vs Numerical Analysis

1. Price by Gender



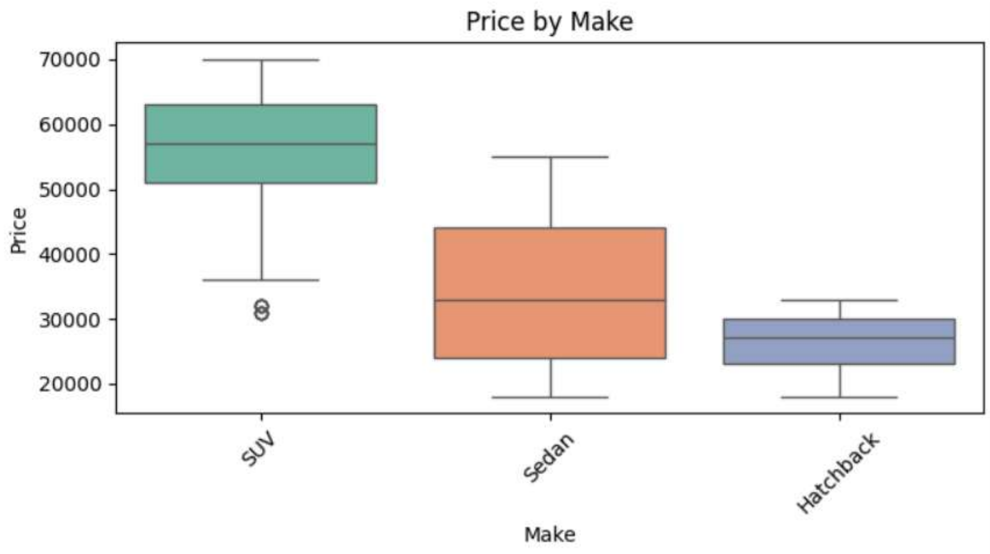
There isn’t a huge difference in car prices between genders, though male buyers show a slightly wider range. This could indicate more varied preferences or purchasing power.

2. Price by Profession



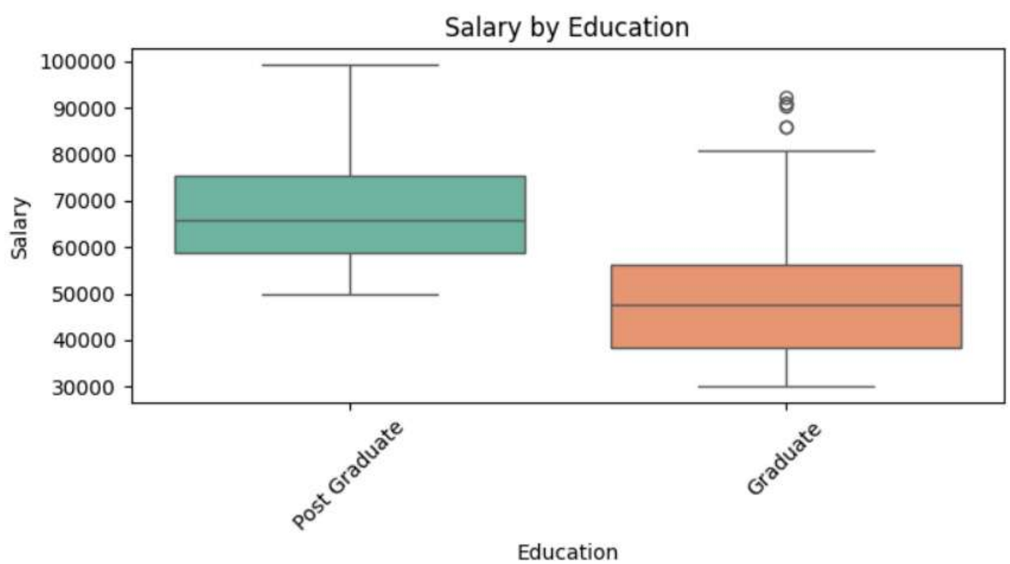
Profession appears to influence car prices to some extent. Business professionals seem to go for a wider price range, while salaried individuals tend to cluster around a slightly lower median.

3. Price by Make



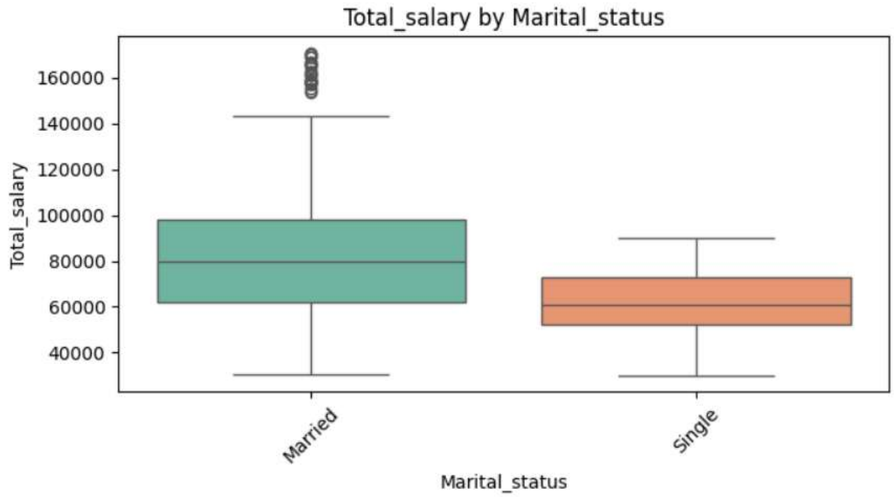
As expected, SUVs and Sedans are associated with higher prices than Hatchbacks. SUVs in particular show a broader spread, suggesting variability in model types and features.

4. Salary by Education



Postgraduates tend to have higher salaries on average compared to graduates. The spread is also slightly wider, reflecting more variance in roles and earning potential.

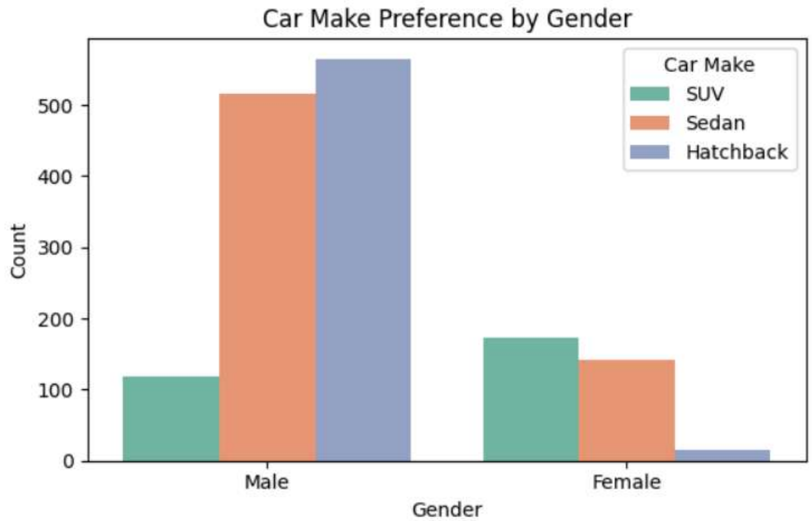
5. Total Salary by Marital Status



Married individuals generally report higher total salaries than single individuals, likely due to dual incomes contributing to household earnings.

INSIGHTFUL QUESTIONS

1. Do men tend to prefer SUVs more compared to women?



The countplot shows that male buyers are fairly evenly split between Hatchbacks and Sedans, with each category having over 500 purchases, while SUV purchases among men are noticeably lower, just above 100. In contrast, female buyers have almost no Hatchback purchases, with Sedans and SUVs both above 120 and 150 respectively. This indicates that while men tend to favor smaller cars like Hatchbacks and Sedans, women show a stronger preference for SUVs compared to men.

2. What is the likelihood of a salaried person buying a Sedan?

Our analysis shows that about 44.2% of salaried individuals purchased Sedans. This indicates that Sedans are a favored choice for salaried buyers, likely because they offer a good mix of comfort and affordability that suits this group's preferences.

3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for an SUV sale over a Sedan sale?

Among salaried male buyers, there were 85 SUV purchases compared to 279 Sedan purchases. This suggests that salaried males actually prefer Sedans over SUVs by a significant margin. Therefore, the data does **not** support the claim that salaried males are easier targets for SUV sales; instead, Sedans remain the dominant choice in this group.

4. How does the amount spent on purchasing automobiles vary by gender?

The average price of cars bought by female buyers is about 47,705, with a median of 49,000, which is significantly higher than male buyers, whose average price is around 32,817 and median price is 29,000. This suggests that, on average, females tend to spend more on purchasing automobiles compared to males in this dataset.

5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

Individuals who took personal loans spent a total of approximately $27,290,000 on purchasing automobiles. This indicates that personal loans might be enabling customers to afford higher-priced vehicles or increase their purchasing power overall.

6. How does having a working partner influence the purchase of higher-priced cars?

The boxplot and summary statistics show that the average prices paid by buyers with working partners (mean ≈ $35,267) and those without (mean ≈ $36,000) are quite similar, and both groups have the same median price of $31,000. This suggests that having a working partner doesn't significantly impact the amount spent on cars in this dataset.

ACTIONABLE INSIGHTS

Based on the data analysis, here are some practical recommendations to help guide marketing and sales strategies:

1. Target Female Buyers for SUVs: Women show a stronger preference for SUVs compared to men. Tailoring SUV promotions, financing options, and advertising toward female customers could help boost sales in this segment.

2. Focus on Sedans for Salaried Customers: Almost half of salaried buyers choose Sedans, making this a key vehicle type for this demographic. Offering attractive deals or financing specifically for Sedans may increase purchases among salaried individuals.

3. Reconsider SUV Campaigns for Salaried Males: Contrary to some expectations, salaried males prefer Sedans far more than SUVs. Marketing efforts for SUVs should be redirected toward segments with a higher interest in SUVs, rather than focusing heavily on salaried males.

4. Leverage Higher Spending by Female Buyers: Female customers tend to spend significantly more on vehicles on average than male customers. Offering premium models, luxury features, or personalized services could capture more value from this group.

5. Promote Financing Options: Individuals with personal loans contribute a large portion of total sales, indicating that financing plays a crucial role in purchasing decisions. Expanding loan offerings or making financing more accessible could help increase overall sales.

6. Consider Individual Buyer Profiles Over Partner Employment Status: Since having a working partner does not significantly influence purchase price, marketing strategies should focus more on individual buyer characteristics rather than household income or partner employment.

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